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Department of Computer Science
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F.Y.B.Sc.(Data Science) Sem – I
Problem Solving and Python Programming
Question Bank

Unit1: Introduction to Problem Solving

1. Define the following terms: compiler, assembler, interpreter, debugger
 2. Define algorithm.
 3. List the characteristics of an algorithm.
 4. Write any two advantages of algorithm.
 5. Write any two limitations of algorithm.
 6. Define flowchart.
 7. What is assembly language?
 8. What is machine language?
 9. Write any two advantages of flowchart.
 10. Write any two limitations of flowchart.
 11. What is low level Language? Write any 2 advantages of it.
 12. What is high level Language? Write any 2 advantages of it.
 13. Write advantages and Limitations of low level language.
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- 1) What is problem solving? And Write Problem solving steps?
 - 2) Define algorithm? Explain its characteristics with proper example?
 - 3) Define Flowchart? Describe notations to draw flowchart with proper example.
 - 4) State advantages and limitations of algorithm?
 - 5) State advantages and limitations of flowchart?
 - 6) Compare algorithm with flowchart.
 - 7) What is Pseudo code? State advantages and limitations of it?
 - 8) What is Programming Language? Explain its type?
 - 9) Practice on drawing algorithm and flowchart.

Unit2: Introduction to Python

- 1) What is indentation?
- 2) What is a slice operator?
- 3) What is comment
- 4) Give the use of index() in string.
- 5) What is the use of if statement.
- 6) What is the use of break statement?
- 7) What is the use of continue statement?
- 8) What is the use of pass statement?
- 9) What is mutable?
- 10) What is immutable?
- 11) What is string?
- 12) What is list?
- 13) What is tuple?
- 14) What is set?
- 15) What is dictionary?
- 16) What is function?
- 17) Write syntax to define a function.
- 18) What is anonymous function?

- 19) List features of Python.
- 20) How are comments written in Python?
- 21) Compare list and tuple (any two points).
- 22) Write code to print the elements of the list `l1 = [10, 20, 30, 40, 50]`
- 23) Explain the range() function and its parameters.
- 24) What is the difference between pop() and del in list.
- 25) How to add multiple elements at the end of the list?
- 26) What is function? Write syntax to define a function.

- 27) Write a note on Python data types.
- 28) Explain slicing concept in python.
- 29) Explain any four string methods.
- 30) Explain while loops with example.
- 31) Explain for loops with example.

- 32) Explain the following statement
a) if b) if else c) while d) for
- 33) How to perform input output operations? Explain with example.
- 34) What are the usage of tuple(), count() and index() functions of tuple?
- 35) What are the usage of dictionary copy(), gets(), items() and keys() methods?
- 36) Explain union, intersection, Difference and Symmetric difference of Set with example.
- 37) Explain syntax to define a function and Call a function with proper example.
- 38) Write a note on function arguments?
- 39) Explain global & local variable with Examples.
- 40) What is anonymous function or lambda function? Explain with example.
- 41) Python programs using functions.

Unit3: Classes, Objects and Inheritance

1. What is class?
2. What is object?
3. What is inheritance?
4. What is hybrid inheritance?
5. What is hierarchical inheritance?

6. What is class? Write syntax to create a class.
7. What is object? Write syntax to create a object.
8. What is inheritance? Write its advantages.
9. Write syntax for single inheritance.
10. What are different kinds of relationships hold by hierarchical inheritance.

11. Demonstrate how to create a class with suitable example?
12. What is inheritance? Explain its types.
13. Explain single, multiple, multilevel and hierarchical inheritance.
14. Python programs using class.
15. Python programs using inheritance.

Unit4: Modules and Packages

1. What is module?
2. List any 4 built in modules used in python.
3. Explain any 4 methods of maths module.
4. What is package? List its types.
5. What is module? Write syntax to create and import a module in python program.
6. What is user defined package. Write syntax to create and import it in python.
7. Write a program to demonstrate to create a module and import it.
8. Write a program to demonstrate to create a user defined package and to import it.